Right-Triangle Trigonometry

ANSWER KEY

Trigonometric ratios

1 A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin\theta \ = \frac{5}{13}$ **b** $\cos\theta \ = \frac{12}{13}$ **c** $\tan\theta \ = \frac{5}{12}$

2 Write the exact value of each expression:

a $\sin 30° \ = \frac{1}{2}$ **b** $\cos 45° \ = \frac{\sqrt{2}}{2}$ **c** $\tan 60° \ = \sqrt{3}$ **d** $\sin 60° \ = \frac{\sqrt{3}}{2}$

3 A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse. $\sqrt{7^2+24^2} = \sqrt{625} = 25$.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.
$\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26°$.

Applications

4 A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.
$h = 4.5\tan 62° \approx 4.5 \times 1.8807 \approx 8.46$ m.

5 From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.
$d = \frac{50}{\tan 18°} \approx \frac{50}{0.3249} \approx 153.9$ m.

6 A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.
$h = 85\sin 54° \approx 85 \times 0.8090 \approx 69$ m.

7 Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is $32°$. Find the height of the taller building, to one decimal place.
Rise $= 40\tan 32° \approx 40(0.6249) \approx 25.0$ m, so the taller building is about $25 + 25.0 = 50.0$ m.

8 A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as $37°$. Find the height of the tower, to the nearest tenth of a metre.
$h = 120\tan 37° \approx 120(0.7536) \approx 90.4$ m.

Solving right triangles

9 In right triangle ABC with the right angle at C, $A = 35°$ and the hypotenuse $AB = 20$. Find:

a side BC (opposite A) $BC = 20\sin 35° \approx 11.47$.

b side AC (adjacent to A) $AC = 20\cos 35° \approx 16.38$.

c angle B $B = 90° - 35° = 55°$.

10 In right triangle PQR with the right angle at R, the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.
$\tan P = \frac{12}{9}$, so $P = \tan^{-1}\left(\frac{12}{9}\right) \approx 53.1°$. Hypotenuse $= \sqrt{9^2 + 12^2} = 15$.

Angles of elevation and depression

11 A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is $27°$. Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.
$d = \frac{60}{\tan 27°} \approx \frac{60}{0.5095} \approx 118$ m.

12 From a lighthouse 45 m tall, the angle of depression to a ship is $12°$, and a second ship is spotted further out with angle of depression $7°$. Find the distance between the two ships, to the nearest metre.
Distances from the base: $d_1 = \frac{45}{\tan 12°} \approx 211.7$ m; $d_2 = \frac{45}{\tan 7°} \approx 366.5$ m. Distance between ships $\approx 366.5 - 211.7 \approx 154.8$ m.

The reciprocal trig ratios

13 For a right triangle with opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$ (relative to angle θ), find:

a $\csc\theta$ $= \frac{5}{3}$ **b** $\sec\theta$ $= \frac{5}{4}$ **c** $\cot\theta$ $= \frac{4}{3}$

14 Explain why $\csc\theta = \frac{1}{\sin\theta}$, and use this to find $\csc(30°)$.
$\csc\theta$ is defined as the reciprocal of $\sin\theta$ (hypotenuse over opposite, instead of opposite over hypotenuse). $\csc(30°) = \frac{1}{\sin(30°)} = \frac{1}{1/2} = 2$.

Multi-step surveying problems

15 A surveyor at point A measures the angle of elevation to a mountain peak as $22°$. Walking 500 m directly toward the mountain to point B, the angle of elevation increases to $35°$. Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)
From B: $h = x\tan 35°$. From A: $h = (x + 500)\tan 22°$. Setting equal: $x\tan 35° = (x + 500)\tan 22°$. $x(0.7002) = (x + 500)(0.4040)$. $0.7002x - 0.4040x = 202$. $0.2962x = 202 \Rightarrow x \approx 682.0$. $h = 682.0(0.7002) \approx 477.5$ m.

The 30-60-90 and 45-45-90 special triangles

16 In a $30\text{-}60\text{-}90$ triangle, the shortest side (opposite the $30°$ angle) has length 5. Find the lengths of the other two sides in exact form.

Side ratios are $1:\sqrt{3}:2$. Side opposite $60°$: $5\sqrt{3}$. Hypotenuse: 10.

17 In a $45\text{-}45\text{-}90$ triangle, each leg has length 7. Find the hypotenuse in exact form.

Hypotenuse $= 7\sqrt{2}$ (using the ratio $1:1:\sqrt{2}$).

Trigonometric ratios beyond the first quadrant (preview)

18 A right triangle used to define $\sin\theta$ and $\cos\theta$ only works directly for acute angles θ. Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.

A right triangle can only have angles strictly between $0°$ and $90°$ (since it already has one $90°$ angle and angles sum to $180°$), so triangle ratios cannot directly define sine or cosine for angles at or beyond $90°$. Extending to all angles requires the unit-circle definition, where $\cos\theta$ and $\sin\theta$ are the x- and y-coordinates of a point on the unit circle at angle θ.